



Question 1: Multiple Choice Question

(35 marks)

Answer:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35					

- Which of the following functions are performed by UDP?
 - process-to-process communication
 - node-to-node communication
 - host-to-host communication
 - None of the choices are correct
- The source port number on the UDP user datagram header defines _____.
 - the sending computer
 - the receiving computer
 - the process running on the sending computer
 - None of the choices are correct
- UDP packets have a fixed-size header of _____ bytes.
 - 16
 - 8
 - 40
 - 32
- TCP groups a number of bytes together into a packet called a _____.
 - user datagram
 - segment
 - datagram
 - None of the choices are correct
- TCP uses _____ to check the safe and sound arrival of data.
 - an acknowledgment mechanism
 - out-of-band signaling
 - the services of another protocol
 - None of the choices are correct
- In TCP, a SYN + ACK segment consumes _____ sequence numbers.
 - no
 - three
 - two
 - one
- A socket address is a combination of _____.
 - A MAC address and a logical address
 - A MAC address and a port number
 - a user-specific address and a port number
 - None of the choices are correct
- _____ means accepting items from more than one source. _____ delivering items to more than one source.
 - Demultiplexing; Multiplexing
 - Multiplexing; Demultiplexing
 - Encapsulation; Decapsulation
 - Pulling; Pushing
- In TCP, if the ACK value is 200, then byte _____ has been received.
 - 199
 - 200
 - 201
 - None of the choices are correct



10. In TCP's _____ algorithm the size of the congestion window increases exponentially until it reaches a threshold.
- congestion avoidance
 - congestion detection
 - slow start
 - None of the choices are correct
11. In the Go-Back-N protocol, the maximum send window size is _____ and the maximum receive window size is _____ where m is related to the number of bits in the sequence number.
- 1, 1
 - 1, 2^m
 - 1, 1
 - 1, 2^m
12. The network layer in the Internet provides _____.
- comprehensive error and flow control
 - limited error control, but no flow control
 - comprehensive error control but limited flow control
 - All of the choices are correct
13. In a datagram approach, the forwarding decision is based on the value of the _____ field in the packet header.
- source address
 - destination address
 - label
 - None of the choices are correct
14. IP is a _____ protocol.
- connection-oriented unreliable
 - connection-oriented reliable
 - connectionless unreliable
 - connectionless reliable
15. If the fragment offset has a value of 200 it means that _____.
- the datagram has not been fragmented
 - the datagram is 200 bytes in size
 - the first byte of the datagram is byte 1600
 - None of the choices are correct
16. What is needed to determine the number of the last byte of a fragment?
- offset number
 - total length
 - both offset number and the total length
 - None of the choices are correct
17. When a datagram is encapsulated in a frame, the total size of the datagram must be less than the _____.
- MUT
 - MAT
 - MTU
 - None of the choices are correct
18. In subnetting, the number of addresses in each subnet must _____.
- be a power of 2
 - be a multiple of 128
 - be divisible by 128
 - None of the choices are correct
19. What is the default prefix length for class C in CIDR notation?
- 24
 - 8
 - 16
 - None of the choices are correct
20. The idea of address aggregation was designed to alleviate the increase in routing table when using _____ addressing.
- classful
 - classless
 - classful or classless
 - None of the choices are correct
21. Which of the following is true about ICMP messages?
- An ICMP error message may be generated for an ICMP error message.
 - An ICMP error message may be generated for a fragmented datagram.
 - An ICMP error message for a multicast datagram.
 - None of the choices are correct.
22. Routing between autonomous systems is referred to as _____ routing.



- A. interdomain
B. intradomain
C. out-of-domain
D. None of the choices are correct
23. A one-to-many communication between one source and a specific group of hosts is classified as a _____ communication.
A. unicast
B. multicast
C. broadcast
D. None of the choices are correct
24. Communication at the data-link layer is _____.
A. end-to-end
B. node-to-node
C. process-to-process
D. None of the choices are correct
25. Which error detection method consists of just one redundant bit per data unit?
A. Two-dimensional parity check
B. CRC
C. Simple parity check
D. Checksum
26. In CRC, if the dataword is 111111 and the divisor 1010, what is the dividend at the sender?
A. 1111110000
B. 111111000
C. 111111
D. 1111111010
27. If the sender is a host and wants to send a packet to another host on another network, the logical address that must be mapped to a physical address is _____.
A. the destination IP address in the datagram header
B. the IP address of the router found in the routing table
C. the source IP address
D. None of the choices are correct
28. In the _____ random-access method, stations do not sense the medium.
A. CSMA/CA
B. ALOHA
C. CSMA/CD
D. Ethernet
29. The sender is a router that has received a datagram destined for a host in the same network. The logical address that must be mapped to a physical address is _____.
A. the destination IP address in the datagram header
B. the IP address of the router found in the routing table
C. the source IP address
D. None of the choices are correct
30. An ARP request is normally _____.
A. broadcast
B. multicast
C. unicast
D. None of the choices are correct
31. Given the Ethernet address 01011010 00010001 01010101 00011000 10101010 11001 binary, what is the address in hexadecimal notation?
A. 5A:88:AA:18:55:FC
B. 5A:81:BA:81:AA:CF
C. 5A:18:5A:18:55:CF
D. 5A:11:55:18:AA:CF
32. The _____ layer of Ethernet consists of the LLC sublayer and the MAC sublayer.
A. data link
B. physical
C. network
D. None of the choices are correct



33. Each station on an Ethernet network has a unique _____ address imprinted on its network interface card (NIC)

A. 16-bit

B. 32-bit

C. 64-bit

D. None of the choices are correct

34. Gigabit Ethernet has a data rate of _____ Mbps.

A. 10

B. 100

C. 1000

D. 10,000

35. A VLAN as a local area network configured by _____

A. software

B. physical wiring

C. software or physical wiring

D. None of the choices are correct

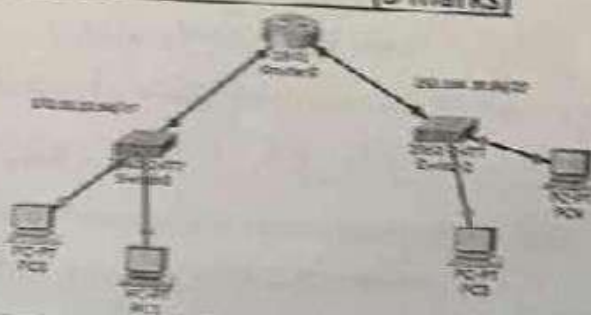
Question 2:

1. What addresses does the PC-3 need?

(5 marks)

2. What is the broadcast address for this network?

3. One of the allocation methods is to dynamically distribute the logical address on the network. How is that done?



Question 3:

1. Write the mac table address for the SW1?

(7 marks)

